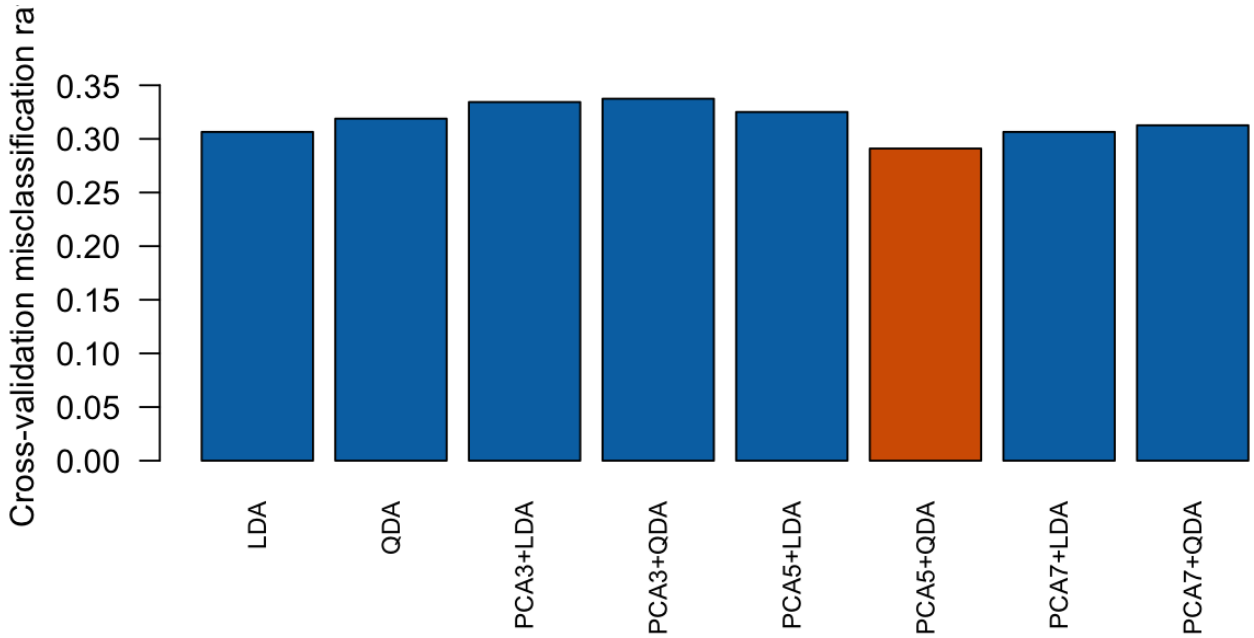
SAheart: LDA/QDA model comparison

Seed 432074 | 70/30 stratified split | Five shared stratified CV folds
Response: chd | Nine numeric predictors; famhist coded Absent=1, Present=2

Candidates and validation results



Model	CV error	Fold SD
LDA	0.307	0.045
QDA	0.319	0.070
PCA3+LDA	0.334	0.040
PCA3+QDA	0.337	0.054
PCA5+LDA	0.325	0.058
PCA5+QDA (selected)	**0.291**	**0.061**
PCA7+LDA	0.307	0.059
PCA7+QDA	0.313	0.047

Selected fit and reserved test set

PCA5+QDA was refitted on all 323 training cases.
Test error: 32/139 = 0.230 (23.0%).
PCA scaling and directions were re-estimated within each CV fitting fold.

Finding and limitation

PCA5+QDA had the lowest CV error. Its test error was lower, but the test set contains only 139 cases; small model differences remain uncertain.